

Utkarsh Bharti

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EXPERIENCE

Honeywell

Jun. 2024 – Aug. 2024

Software Engineering Intern (ML Systems)

- Engineered **high-performance surrogate model pipelines** in **Python (Scikit-learn, TensorFlow)** that replaced expensive CFD simulation cycles, cutting compute time by **30–40%** and accelerating iteration speed
- Designed and implemented a **modular data preprocessing pipeline** using **NumPy and Pandas** to handle high-dimensional turbine geometry datasets, reducing feature space by **40%** via dimensionality reduction
- Built an **automated optimization backend** using reinforcement learning to iteratively tune aerodynamic parameters under engineering constraints, eliminating manual simulation overhead
- Developed a **performance monitoring and visualization module** tracking **15+ turbine metrics** using **Matplotlib and Seaborn**, enabling data-driven engineering decisions
- Ranked 3rd among 100+ teams** in the Honeywell Designathon and awarded **Rs. 25,000** for engineering impact

Zephyra Dynamics

Jan. 2024 – May 2024

Software Engineering Intern (Backend & Data Systems)

- Architected an **end-to-end fault detection backend system** in **Python (Scikit-learn, Pandas)** ingesting heterogeneous sensor streams and automating anomaly triage with zero manual intervention
- Engineered a **scalable data ingestion and preprocessing pipeline** processing **10,000+ time-series sensor records**, applying sliding-window transformations and signal-based feature extraction
- Built and evaluated **classification and anomaly detection models** for predictive diagnostics, achieving **18–22% improvement** in fault detection accuracy over the existing rule-based baseline
- Designed **signal-derived feature engineering** logic that extracted temporal and frequency-domain patterns, directly improving downstream model precision
- Delivered a **complete ML inference pipeline** covering data ingestion, preprocessing, model serving, and structured performance reporting for production handoff

PROJECTS

Resume Analyzer | *Python, FastAPI, spaCy, Pandas, REST APIs* | [GitHub](#)

- Built a **production-grade resume analysis service** exposing **REST APIs** that parse resumes against job descriptions and return structured improvement reports, supporting **concurrent multi-user requests**
- Engineered an **NLP extraction pipeline** using **spaCy** to identify skills, experience spans, and keywords from unstructured resume text with high precision
- Designed a **weighted keyword scoring engine** that computes ATS compatibility scores by comparing extracted entities against JD requirements using inverted-index lookups for **O(1) term matching**
- Structured the backend with **clean separation of concerns** ingestion, parsing, scoring, and reporting layers enabling modular testing and easy feature extension

NexChat Real-Time Chat Application | *Node.js, React.js, WebSocket, PostgreSQL, REST APIs* | [GitHub](#)

- Developed a **full-stack real-time messaging platform** handling **multi-user concurrent sessions** via **WebSocket connections**, achieving sub-100ms message delivery latency
- Designed a **scalable REST API layer** for user authentication, session management, and message history retrieval backed by **PostgreSQL** with indexed queries for fast lookups
- Implemented **event-driven message broadcasting** on the server using pub/sub patterns to efficiently route messages across active connections without polling overhead
- Optimized **high-concurrency message handling** by decoupling WebSocket I/O from database writes, reducing response latency under load and improving system throughput

AI Automated Troubleshooting System | *Python, Scikit-learn, Pandas, NumPy, Matplotlib* | [GitHub](#)

- Engineered a **fault diagnosis backend platform** that ingests large-scale heterogeneous sensor datasets and automatically identifies anomalies and system failure patterns
- Built a **robust data pipeline** handling **structured and unstructured sensor records**, performing noise filtering, normalization, and feature extraction prior to model inference
- Implemented **classification and root-cause analysis models** for predictive diagnostics, packaged as a reusable inference module with configurable thresholds
- Delivered **end-to-end system coverage** from raw data ingestion and preprocessing through model inference to **structured reporting dashboards** enabling actionable insights with no manual analysis

EDUCATION

R.V College of Engineering

Nov. 2022 – Jul. 2026

Bachelor of Engineering in Civil Engineering; Minor in Artificial Intelligence

Bengaluru, Karnataka

- Relevant Coursework:** Data Structures & Algorithms, Databases, Operating Systems, Web Development, Machine Learning, Computer Networks, GIS/Remote Sensing, AutoCAD, Revit

ACHIEVEMENTS

Competitive & Hackathon Recognition

- 2nd Place HAL Hackathon:** Secured runner-up position among competing teams at the Hindustan Aeronautics Limited Hackathon
- 7th Rank DSA Competition by AlgoUniversity:** Ranked in the top finishers in a college-wide Data Structures & Algorithms contest
- 3rd Place Honeywell Designathon:** Ranked 3rd among 100+ teams and awarded **Rs. 25,000** for engineering impact
- NASA Space Apps Challenge Participant:** Contributed to a globally recognized open-data hackathon, developing solutions to real-world space and Earth science challenges

CERTIFICATIONS

AI & Machine Learning

- NVIDIA Fundamentals of Deep Learning:** Hands-on training in neural network architectures, model training workflows, and GPU-accelerated deep learning using NVIDIA's DLI platform
- Microsoft Machine Learning Fundamentals:** Covered supervised/unsupervised learning, model evaluation, and ML pipeline concepts on Microsoft Learn

Data & Analytics

- Google Google Analytics Certification:** Certified in web analytics, traffic analysis, conversion tracking, and data-driven decision making using Google Analytics

SKILLS

Languages : Python, JavaScript (ES6+), SQL (PostgreSQL, MySQL), HTML/CSS

Frameworks & Libraries : React.js, Node.js, FastAPI, Scikit-learn, TensorFlow, Pandas, NumPy, spaCy

Core CS : Data Structures & Algorithms, OOP, REST APIs, WebSockets, System Design Fundamentals, OS Basics

Tools : Git, GitHub, Postman, Figma, Jira, Notion, Google Analytics, DaVinci Resolve, OBS

Platforms & Certifications : NVIDIA Deep Learning Institute, Microsoft Learn, Google Analytics, NASA Space Apps